Analysis of 2016 Hate Crime Database

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Loading required packages:

Loading data

Let's load data from .csv file obtained from FiveThirtyEight.

Reference: ("https://raw.githubusercontent.com/fivethirtyeight/data/master/hate-crimes/hate_crimes.c
sv"):

```
hate_crimes <- read.csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/hate-crimes/hate_
glimpse(hate_crimes)
```

```
## Rows: 51
## Columns: 12
## $ state                                  <chr> "Alabama", "Alaska", "Arizona~
## $ median_household_income                <int> 42278, 67629, 49254, 44922, 6~
## $ share_unemployed_seasonal              <dbl> 0.060, 0.064, 0.063, 0.052, 0~
## $ share_population_in_metro_areas        <dbl> 0.64, 0.63, 0.90, 0.69, 0.97,~
## $ share_population_with_high_school_degree <dbl> 0.821, 0.914, 0.842, 0.824, 0~
## $ share_non_citizen                      <dbl> 0.02, 0.04, 0.10, 0.04, 0.13,~
## $ share_white_poverty                    <dbl> 0.12, 0.06, 0.09, 0.12, 0.09,~
## $ gini_index                             <dbl> 0.472, 0.422, 0.455, 0.458, 0~
## $ share_non_white                        <dbl> 0.35, 0.42, 0.49, 0.26, 0.61,~
## $ share_voters_voted_trump               <dbl> 0.63, 0.53, 0.50, 0.60, 0.33,~
## $ hate_crimes_per_100k_splc              <dbl> 0.12583893, 0.14374012, 0.225~
## $ avg_hatecrimes_per_100k_fbi            <dbl> 1.8064105, 1.6567001, 3.41392~
```

We can see that this dataset has 51 observation points, and 12 variables.

Correlation analysis.

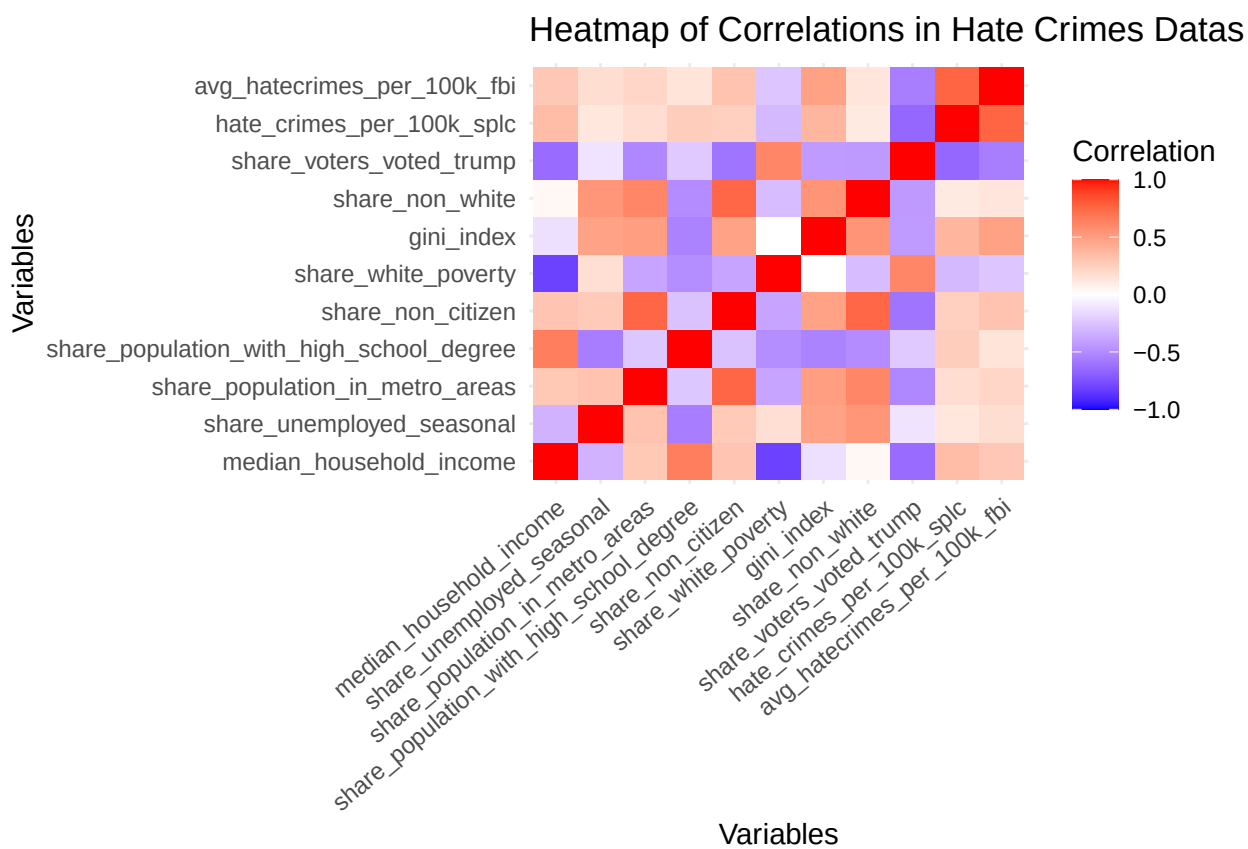
Let's create correlations heatmap for numeric variables, so we can exclude state and state abbreviation variables .

```
numeric_data <- hate_crimes |>
  select_if(is.numeric)
```

```
corr_hate_crimes <- cor(numeric_data, use = "complete.obs")
```

```
melted_hate_crimes <- melt(corr_hate_crimes)
```

```
ggplot(melted_hate_crimes, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  scale_fill_gradient2(low = "blue", high = "red", mid = "white",
                       midpoint = 0, limit = c(-1,1), space = "Lab",
                       name="Correlation") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 40, vjust = 1, hjust = 1)) +
  labs(title = "Heatmap of Correlations in Hate Crimes Dataset",
       x = "Variables",
       y = "Variables")
```



Correlation values closer to 1 show positive correlation, and closer to -1 indicate negative correlation.

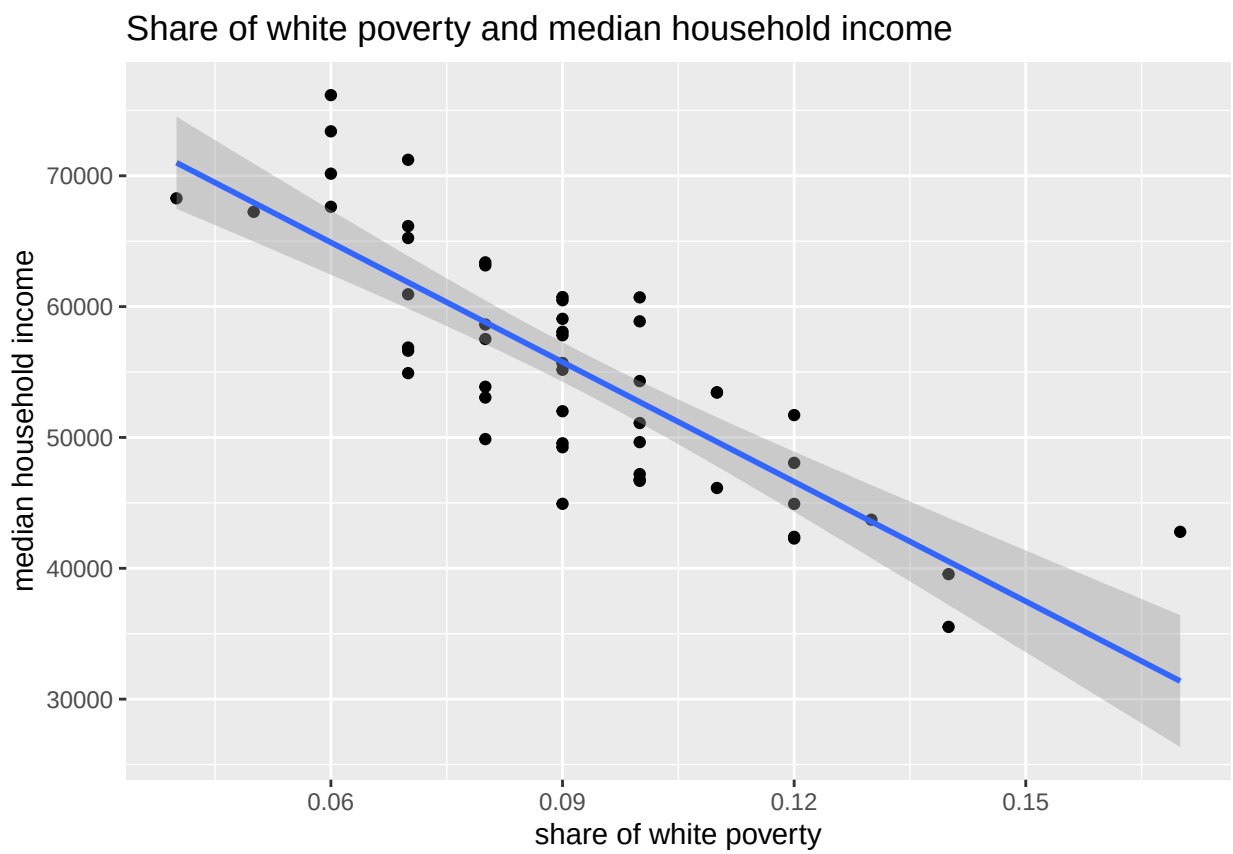
Values close to 0 indicate that there is no significant correlation.

Let's explore some strong positive and negative correlations

Including Plots

1. Let's make a scatter plot to see relationship between share of white poverty and median household income

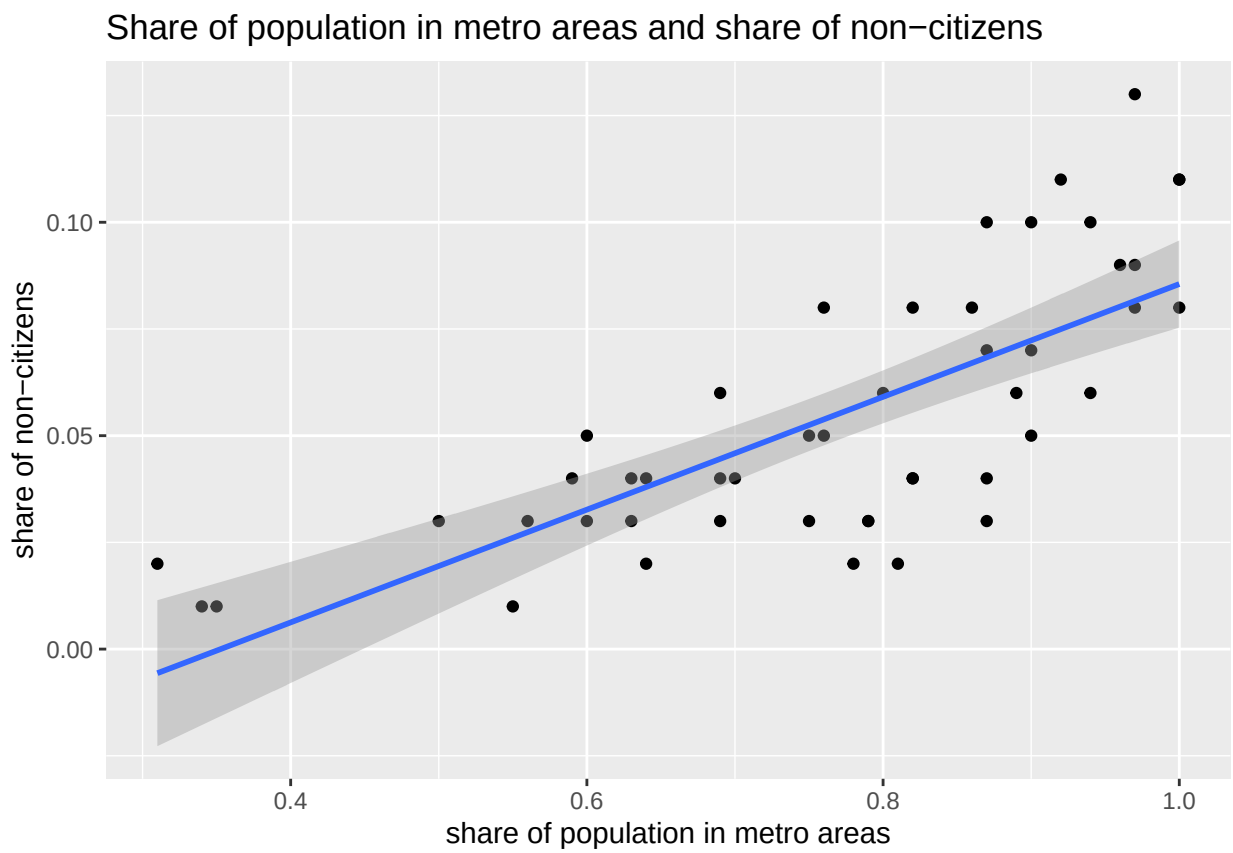
```
ggplot(hate_crimes, aes(x = share_white_poverty, y = median_household_income)) +
  geom_point() +
  geom_smooth(method = "lm")+
  labs(
    title = "Share of white poverty and median household income",
    x = "share of white poverty",
    y = "median household income"
  )
```



The conclusion from this scatter plot is that increase in share of white poverty (explanatory variable) has a negative effect on median household income (response variable).

2. Let's compare share of population in metro areas with share of non-citizens:

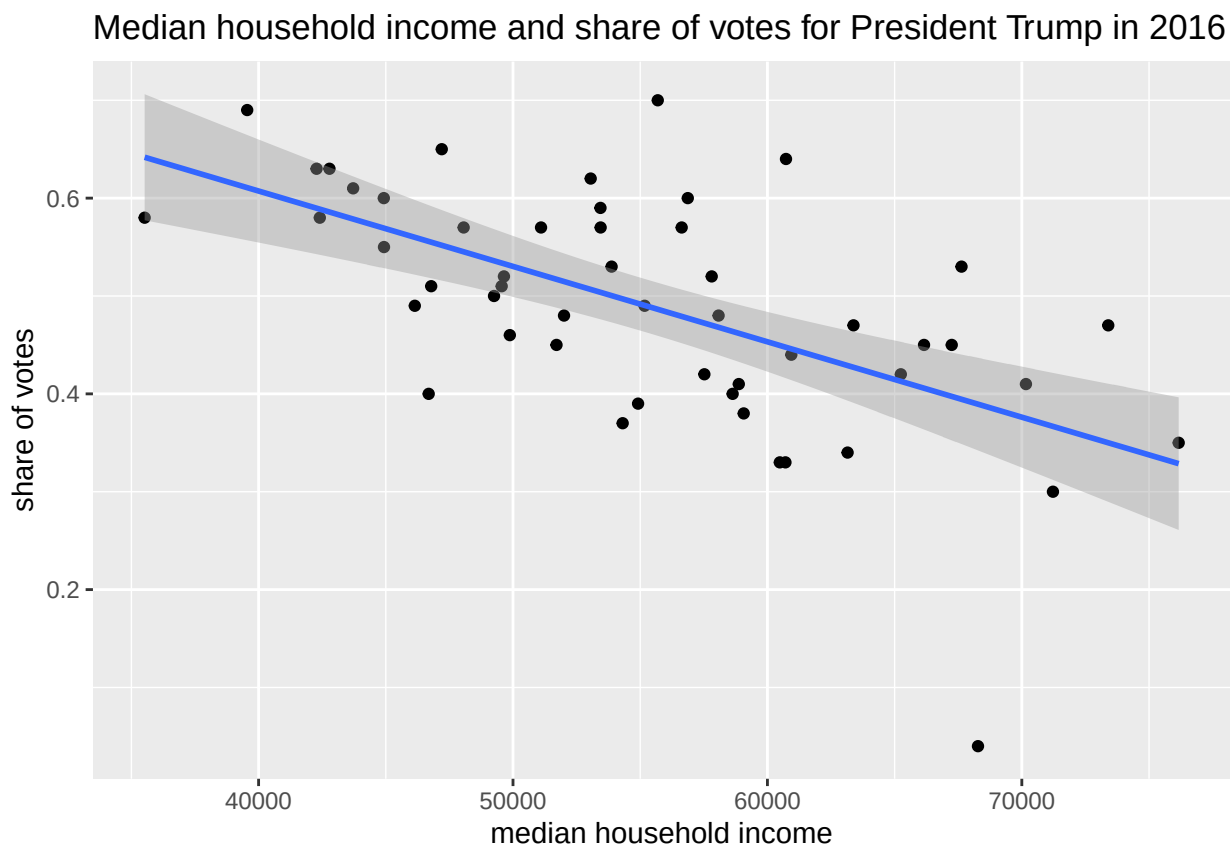
```
ggplot(numeric_data, aes(x = share_population_in_metro_areas, y = share_non_citizen)) +
  geom_point(na.rm = TRUE)+
  geom_smooth(method = "lm") +
  labs(
    title = "Share of population in metro areas and share of non-citizens",
    x = "share of population in metro areas",
    y = "share of non-citizens"
  )
```



In this scatter plot we can see that there is a positive relationship in share of population in metro area (explanatory) and share of non-citizens (response).

3. Let's see how median household income related to share of voters voted for President Trump in 2016:

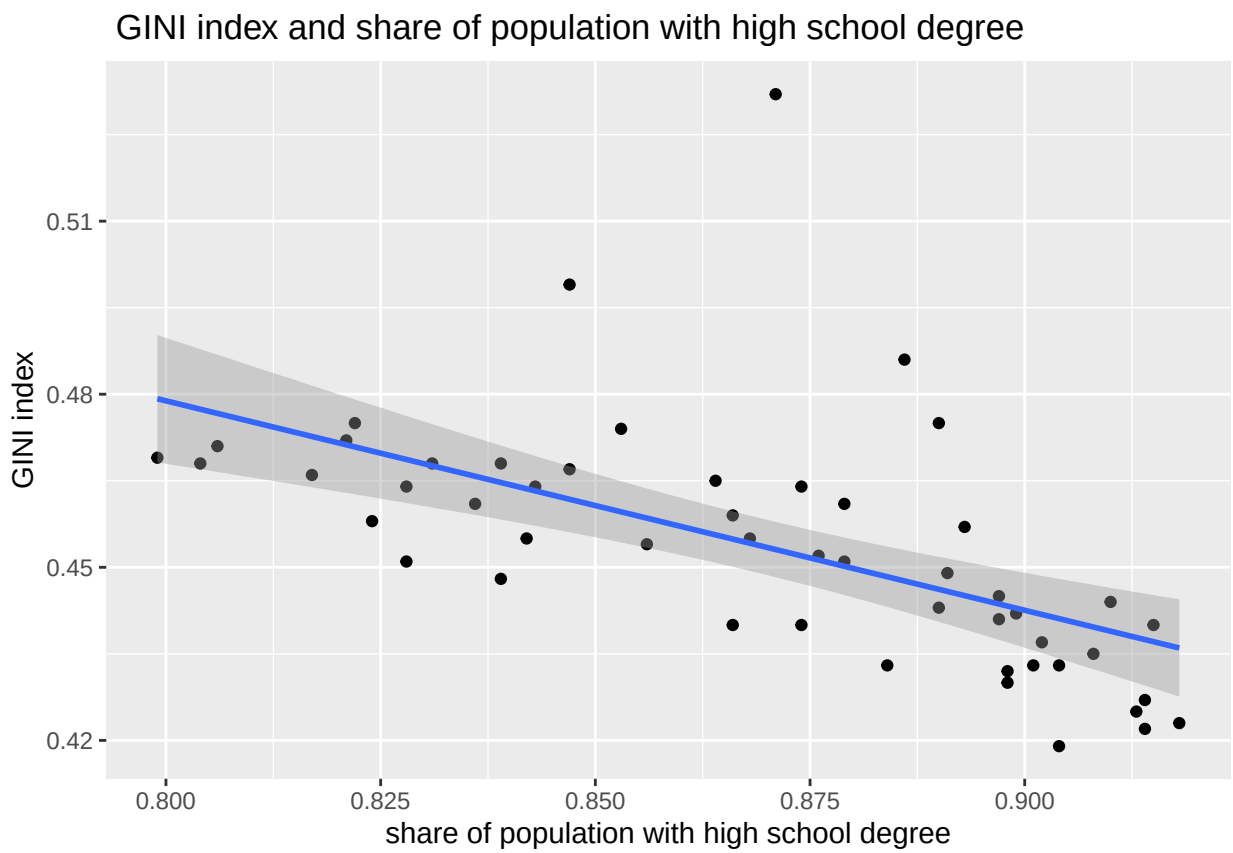
```
ggplot(hate_crimes, aes(x = median_household_income, y = share_voters_voted_trump)) +
  geom_point(na.rm = TRUE) +
  geom_smooth(method = "lm") +
  labs(
    title = "Median household income and share of votes for President Trump in 2016",
    x = "median household income",
    y = "share of votes"
  )
```



Here we can see that increase in median household income (explanatory variable) had a negative effect on share of voters voted for President Trump (response variable) in 2016.

4. Let's see relationship between GINI index and share of population with high school degree:

```
ggplot(hate_crimes, aes(x = share_population_with_high_school_degree, y = gini_index)) +
  geom_point(na.rm = TRUE) +
  geom_smooth(method = "lm") +
  labs(
    title = " GINI index and share of population with high school degree",
    x = "share of population with high school degree",
    y = "GINI index"
  )
```

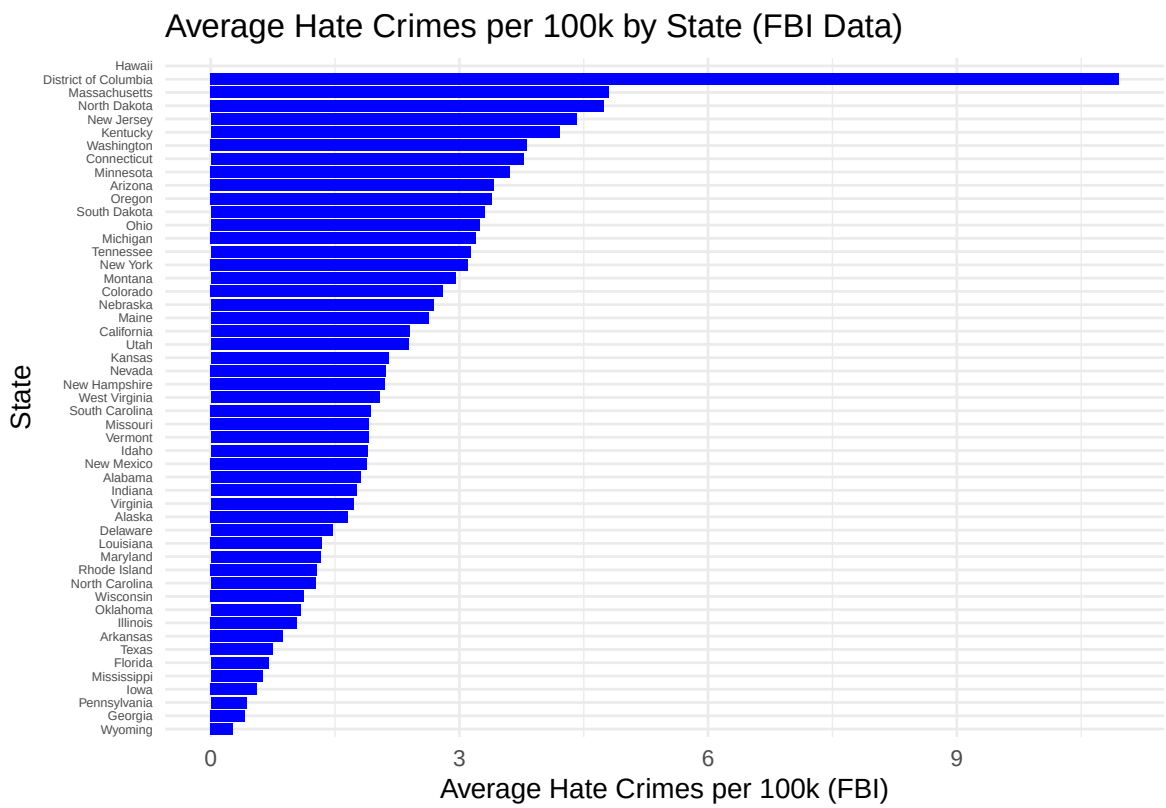


The GINI index is used to measure income inequality, wealth inequality, or consumption inequality. A GINI index of 0 represents perfect equality, while an index of 1 implies perfect inequality.

We can see that increase in share of population with high school diploma (explanatory variable) has a negative impact on GINI index (response variable).

5. Let's create a bar chart for each state count of hate crime for 100K of population per FBI.

```
hate_crimes <- hate_crimes |>
  arrange(avg_hatecrimes_per_100k_fbi)
ggplot(hate_crimes,
       aes(
  x = avg_hatecrimes_per_100k_fbi,
  y = reorder(state, avg_hatecrimes_per_100k_fbi))
  ) +
  geom_col(fill = "blue") +
  labs(
    x = "Average Hate Crimes per 100k (FBI)",
    y = "State",
    title = "Average Hate Crimes per 100k by State (FBI Data)"
  ) +
  theme_minimal() +
  theme(
    axis.text.y = element_text(size = 5, lineheight = 20)
  )
```



This bar chart shows average hate crimes per 100K of population in each state. This data came from FBI.

According to this article https://fivethirtyeight.com/features/higher-rates-of-hate-crimes-are-tied-to-income-inequality/, not all hate crimes included in FBI data. For example, non-prosecuted hate crimes are usually not included in FBI data. Also, many people do not report hate crimes. And we can make a conclusion that this data from FBI is not validated and is not reliable.

Conclusions:

- There are some associated and independent variables in this dataset.
- Education level, race has significant effects on income of population and GINI index.
- Lower income areas had higher votes count for President Trump in 2016.
- The FBI's hate crime data is not validated and is not reliable.
- This dataset could have more associated information, such as categories of education level (some college degrees)